Q No 1 (a) Newton's divided difference formula

$$P_n(x) = f[x_0] + (x-x_0)f[x_0,x_1] + (x-x_0)(x-x_1)f[x_0,x_1,x_2] \cdots$$

$$f[x_0] = f(x_0)$$

$$f[x_0,x_1] = \frac{f(x_1)-f(x_0)}{x_1-x_0}, \quad f[x_1,x_2] = \frac{f(x_2)-f(x_1)}{x_2-x_1}$$

$$f[x_0,x_1,x_2] = \frac{f[x_1,x_2]-f[x_0,x_1]}{x_2-x_0}$$

i	x_i	$f(x_i)$	$f[x_i,x_{i+1}]$	$f[x_i,x_{i+1},x_{i+2}]$	$f[x_i,x_{i+1},x_{i+2},x_{i+3}]$
0	1	3	$\frac{6-3}{2-1}=3$	$\frac{3-3}{4-1}=0$	$\frac{0.2-0}{7-1}$
1	2	6	$\frac{12-6}{4-2}=3$	$\frac{4-3}{7-2}=\frac{1}{5}=0.2$	
2	4	12	$\frac{24-12}{7-4}=4$		
3	7	24			

$$P(x) = 3 + 3(x-x_0) + 0\cdot(x-1)(x-2) + \frac{1}{30}\cdot(x-1)(x-2)(x-4)$$

$\Longleftrightarrow$

Q No 1(b)

$$f(x) = f_0 + \gamma\Delta f_0 + \frac{\gamma(\gamma-1)}{2!}\Delta^2 f_0 + \frac{\gamma(\gamma-1)(\gamma-2)}{3!}\Delta^3 f_0 + \cdots$$

$$\gamma = \frac{x-x_i}{h} \qquad \text{for } x=1.5, \quad \gamma = \frac{1.5-1.0}{0.2} = 2.5$$

i	x_i	$f(x_i)$	$\Delta^1 f_i$	$\Delta^2 f_i$	$\Delta^3 f_i$
0	1.0	2.0			
			$f(x_1)-f(x_0)=0.303$		
1	1.2	2.303		$0.333-0.303=0.033$	
			$f(x_2)-f(x_1)=0.336$		-0.008
2	1.4	2.639		$0.361-0.336=0.025$	
			$f(x_3)-f(x_2)=0.361$		
3	1.6	3.0			

$$f(1.5) = 2 + (2.5)(0.303) + \frac{(2.5)(1.5)}{2}(0.033) + \frac{(2.5)(1.5)(0.5)}{6}(-0.008)$$

$$f(1.5) = 2.816875$$

Q No 2

(a) Hermite interpolation is an interpolating method which not only use function $f(x)$ values but also its derivative $f'(x)$ values to produce a polynomial that fits both the values and the slope of the function at given points. Derivative values are included to ensure the less oscilation and the smooth and more accurate fit polynomial.
In real world example, a position and velocity (derivative) data information of moving object.

b) Runge's phenomenon describe the oscillations at the edge intervels of high degree polynomials. With more point the oscillation at boundaries might increases.
Spline helps to fit low degree polynomials

piecewise between data point which prevents Runge's phenomenon.

c) For n+1 data point, there are n intervals and 3 unknowns required for each quadratic equation, so total of 3n unknowns.
2 equations that must pass through the end points. (2n-2) equations for interior points and n-1 equations for first derivatives that must be equal at the interior point and one additional boundary condition.

$$2 + (2n-2) + (n-1) + 1 = 3n$$

d) Numerical Interpolation:

Estimating function values at point between known data points and can be used when discrete data points, need values between measured points, function is unknown and only discrete or tabulated data is available.

Numerical Integration

Estimating the integral when analytical integration is difficult or complex. It is used when the function has no known integral and required to estimate from the discrete data points. The analytical solution is complex or unknown to solve.

Numerical Differentiation

To approximate the derivatives $f'(x)$ or $f''(x)$ from discrete data points.

Q3 (a)

$$\int_a^b f(x)dx = \int_{-1}^{1} f(t)\,dt = \sum_{i=1}^{n} w_i f(t_i)$$

$$\int_{-1}^{1} 5e^x dx = \int_{-1}^{1} f(x)dx = [w_1 f(x_1) + w_2 f(x_2)]$$

$w_1 = w_2 = 1$ $\quad f(x_1) = f\left(-\frac{1}{\sqrt{3}}\right)$

$f(x_2) = f\left(\frac{1}{\sqrt{3}}\right)$

$$f\left(-\frac{1}{\sqrt{3}}\right) = 5e^{-1/\sqrt{3}} =$$

$$f\left(\frac{1}{\sqrt{3}}\right) = 5e^{1/\sqrt{3}} =$$

$$\int_{-1}^{1} 5e^x dx = 5\left(e^{-1/\sqrt{3}} + e^{1/\sqrt{3}}\right)$$

$$= 5(0.561384 + 1.781312)$$

$$= 11.71348$$

Exact Value:

$$\int_{-1}^{1} 5e^x dx = 5e^x \Big|_{-1}^{1} = 5(e - e^{-1})$$

$$= 5(2.718282 - 0.367879)$$

$$= 11.75201$$

Error = Exact − Estimate

$$= 11.75201 - 11.71348$$

$$= 3.85 \times 10^{-2}$$

Accurate upto two decimal places

Q3(b)

$h = 0.25$

$x_0 = 0$
$x_1 = 0.25$
$x_2 = 0.5$
$x_3 = 0.75$
$x_4 = 1.0$
$x_5 = 1.25$
$x_6 = 1.5$
$x_7 = 1.75$
$x_8 = 2$

Formula:-

$$\frac{h}{3}\left[f_0 + 4f_1 + 2f_2 + 4f_3 + 2f\right.$$
$$\left. + 4f_5 + 2f_6 + 4f_7 + f_8\right.$$

$$\frac{0.25}{3}\left[23.67346\right]$$

Q No 4(a) Unmarked Question

Q No 4(b) $x_0=1, x_1=2, x_2=3, x_3=4, x_4=6$

$f(x) = e^x$

$P_{1,2,4}(x) = f(x_1)L_1(x) + f(x_2)L_2(x) + f(x_4)L_4(x)$

$P_{1,2,4}(5) = f(x_1)L_1(5) + f(x_2)L_2(5) + f(x_4)L_4(x)$

$L_1(5) = \frac{5-3}{2-3} \cdot \frac{5-6}{2-6} = -\frac{1}{2}$

$L_2(5) = \frac{(5-2)(5-6)}{3-2 \quad 3-6} = 1$

$L_3(5) = \frac{(5-2)(5-3)}{(6-2)(6-3)} = \frac{1}{2}$

$P_{1,2,4}(5) = e^2 \cdot \left(\frac{-1}{2}\right) + e^3 \cdot 1 + e^6 \cdot \frac{1}{2}$

$= 218.1054$

Q5 (a)

$$\int_{10}^{16} f(x)\,dx$$

$a=10 \quad b=16 \quad b-a = 6 \quad n=6$

$h = 1$

$x_0 = 10$	$f_0 = 1.02$	$h=1$
$x_1 = 11$	$f_1 = 0.94$	
$x_2 = 12$	$f_2 = 0.89$	
$x_3 = 13$	$f_3 = 0.79$	
$x_4 = 14$	$f_4 = 0.71$	
$x_5 = 15$	$f_5 = 0.62$	
$x_6 = 16$	$f_6 = 0.55$	

$$\int_{10}^{16} f(x)\,dx = \frac{3h}{8}\left[f_0 + 3(f_1+f_2+f_4+f_5) + 2f_3 + f_6\right]$$

$$= \frac{3}{8}(12.63) = 4.73825$$

Q5 (b)

$h = 0.1$

$$f'(x+h) = \frac{f(x+h) - f(x)}{h}$$

$f'(1.1) = 19.9817$

$f'(1.2) = 24.4058$

$f'(1.3) = 29.8091$

$f'(1.4) =$ $f(1.5)$ is not exist.